NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY — 9648/02

Paper 2 Core Paper — **September 2011**

Additional Materials: Answer Paper — **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This document consists of **15** printed pages and **1** blank page.

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For Examiner's Use

Section A

Answer **all** the questions in this section.

1 In the search for new biofuels, research has been done into the digestion of wood waste by fungi. The cellulase enzymes produced by the fungi break cellulose into sugars. These sugars can then be converted into ethanol, a biofuel. **Fig. 1.1** shows the stages in this digestion process.

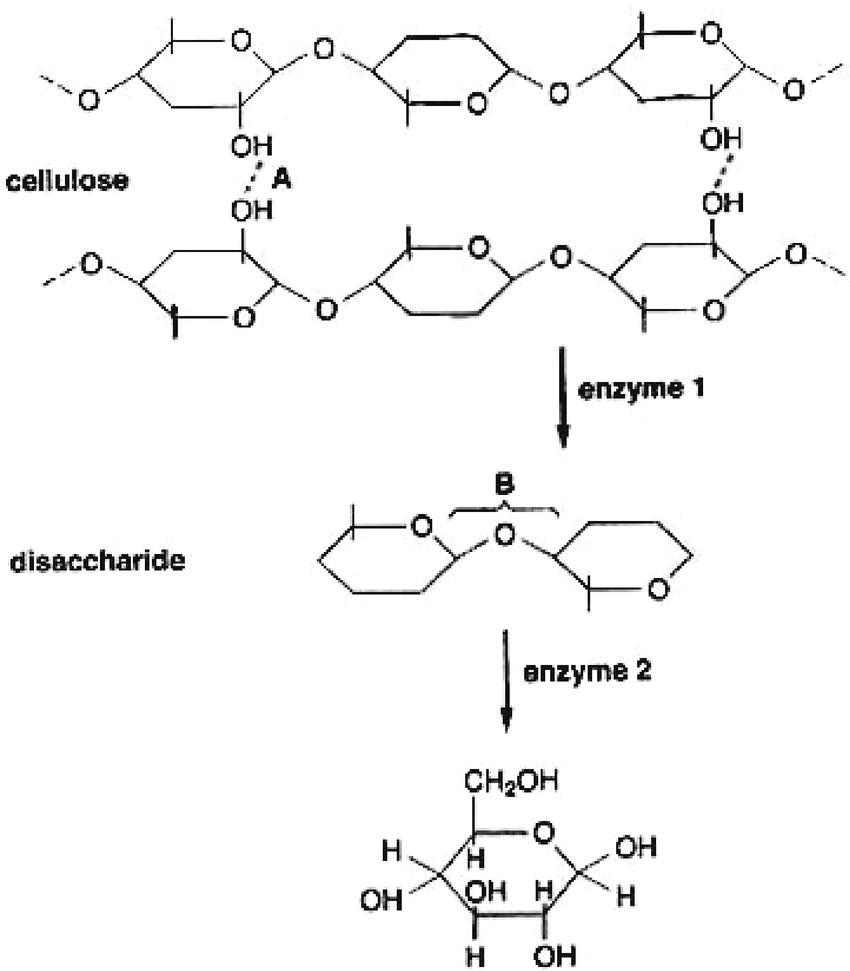


Fig. 1.1

(a) Name bonds **A** and **B** shown in **Fig. 1.1**.

hydrogen bonds;

β(1→4**)** glycosidic bond/ β-1,4 glycosidic bond ®**:** glycosidic bond;

[2]

(b) Describe how bond **B** is broken in the digestion of the disaccharide.

hydrolysis + addition of water;

[1]

(c) Explain why different enzymes are involved in each stage of the digestion process.

different substrates are available at each stage of digestion;

enzymes have specific active sites **complementary** in terms of shape, size, charge to the specific carbohydrate substrate;

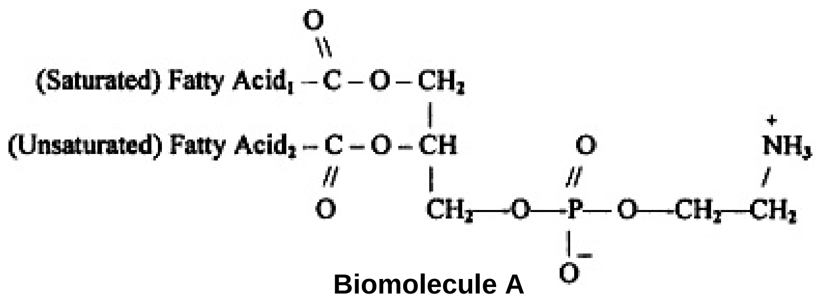
formation of **enzyme-substrate complex**;

ref to lock-and-key/ induced fit hypothesis;

® recognizing different bonds/ active site complementary to bonds

[2]

Fig 1.2 shows a molecules commonly found in cell membranes.



Biomolecule A

Fig. 1.2

(d) Describe the structure of molecule **A Fig. 1.2**.

phosopholipid;

phosphate group and 2 **fatty acid** chains attached to **glycerol**;

Ester bond;

[2]

(e) Describe how the structure of the molecule **A** contributes to its function in membranes.

hydrophilic **phosphate** head interacts with aqueous medium inside and outside the cell;

hydrophobic **hydrocarbon** tails interact with each other forming the hydrophobic core which repels charged/ hydrophilic molecules from passing through;,

unsaturated FA within the tails possess kinks which prevent close packing of PL thus increasing permeability of membrane to non-polar molecules;

[3]

[Total: 10]

2 **Fig 2.1** shows part of a molecule found in a species of bacterium.

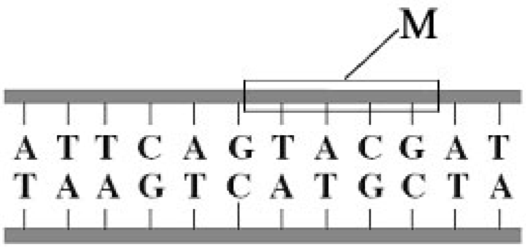


Fig. 2.1

(a) Name the **two** components that make up the region labeled **M**.

deoxyribose sugar;

phosphate; ® **phosphate head**

[2]

(b) Scientists calculated the percentage of different bases in the DNA from a species of bacterium. They found that 14% of the bases were guanine.

(i) What percentage of the bases in this species of bacterium was adenine?

36;

[1]

(ii) In another species of bacterium, 29% of the bases were guanine.

Explain the difference in the percentage of guanine bases in the two species of bacterium.

different proteins are produced;

different genes/ different DNA sequences; ® **mutation / selection pressures**

@ reference to idea that different species have difference genetic makeup

[2]

Fig. 2.2, shows a DNA base sequence. It also shows the effect of two mutations on this base sequence. **Fig. 2.2** shows DNA triplets that code for different amino acids.

A

Original DNA base sequence	A	T	T	G	G	C	G	T	G	T	C	T
Amino acid sequence												
Mutation 1 DNA base sequence	A	T	T	G	G	A	G	T	G	T	C	T
Mutatuib 2 DNA base sequence	A	T	T	G	G	C	C	T	G	T	C	T

B

DNA triplets	**Amino acid**
GGT, GGC, GGA, GGG	Gly
GGT, GTA, GTG, GTC	Val
ATC, ATT, ATA	Ile

TCC, TCT, TCA, TCG	Ser
CTC, CTT, CTA, CTG	Leu

Fig. 2.2

(c) Complete **Fig. 2.2A** to show the sequence of amino acids coded for by the original base sequence.

Ile Gly Val Ser;

[1]

(d) Some mutations affect the amino acid sequences while others do not. Using the information in **Fig. 2.2**, explain

(i) why mutation **1** has no effect on the amino acid sequence;

degeneracy of the genetic code;

® **wobble effect**

[1]

(ii) how mutation **2** could lead to the formation of a non-functional enzyme.

leu replaces val/ change in amino acid sequence/ primary structure;

changes in **R-groups** thus change in hydrogen/ ionic bonds;

alters ® **change** **specific/ precise 3D structure** thus distorts active site;

active site no longer complementary to substrate, denaturation, no E-S complex formed;

[3]

[Total: 10]

3 Cervical cancer occurs in the cervix. The cervix is the opening of the uterus. Scientists investigated the link between cervical cancer and infection with some types of Human Papilloma Virus (HPV). **Fig. 3.1** shows the frequency of five different types of HPV in women who had cervical cancer.

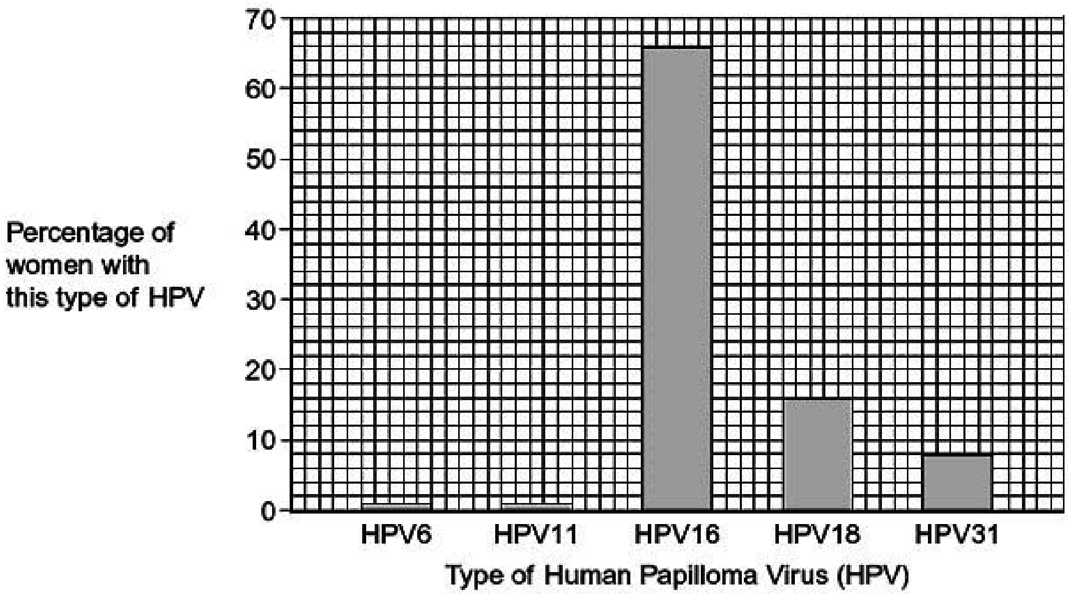


Fig. 3.1

(a) A local newspaper published an article about cervical cancer with the headline "HPV causes cervical cancer."

Using data from **Fig 3.1**, comment if this claim may be supported.

yes + 66% women have HPV 16; **OR**

note: students will only get 2 full marks if they comment that it is NOT supported.

no + only 1% women with cancer has HPV 6/11;

8% of women with cancer did not have HPV ***OR*** no control group ***OR*** HPV in healthy women not studied ***OR*** individuals may also possess more than 1 type of HPV;

note: if no stand taken = 0m ; if both given = max 1m ;

[2]

A study published in the 28th edition of *AIDS* has shown that HIV-positive women are more likely to carry multiple strains of human papilloma virus (HPV). It has been proven that early intervention by antiviral drugs can prevent HIV.

(b) Suggest an explanation for the high incidence of HPV in HIV patients.

HIV infects T cells;

immune system compromised/ not effective;

[2]

(c) Suggest and explain the possible modes of action of the antiviral drugs in halting the reproductive cycle of HIV.

drugs can bind to glycoproteins/ receptor molecules on CD4;
prevent gp120 on HIV viral envelope from binding and entering cell;

drugs can bind to reverse transcriptase;
prevent the synthesis of a single DNA strand that is complementary to the viral RNA;

drugs can bind to/ inhibit integrase;
prevent the dsviral DNA from integrating as provirus into host cell chromosome;

drugs bind to/ inhibit HIV protease;
resulting in viral proteins not being cleaved properly;

[3]

Species **X** and **Y** are bacteria. **Fig. 3.2** shows gene transfer between bacteria in these two species. The bacteria that are shaded are resistant to the antibiotic penicillin.

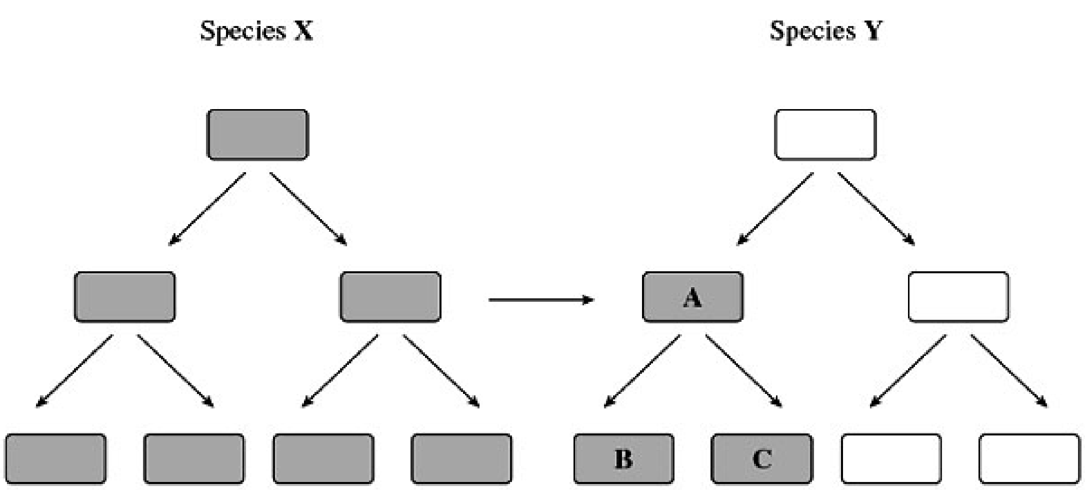


Fig. 3.2

(d) Using **Fig 3.2**,

(i) explain how bacterium **A** becomes resistant to penicillin;

1. conjugation;
2. direct physical interaction between 2 bacteria cells via the **sex pilus**;
3. transfer of penicillin resistance **gene**;
4. from species X to species Y/ Cell A;
5. via **cytoplasmic bridge**
6. ref to plasmid / homologous recombination into host genome;

[4]

For Examiner's Use

(ii) explain why bacteria **B** and **C** are resistant to penicillin.

a) binary fission;

b) replication via semi-conservative **replication**;

c) formation of cells that are **genetically identical to parent cells** after cell division;

[3]

[Total: 14]

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4 Sex hormone estrogen promotes mammary epithelial cell proliferation by stimulating the expression of genes encoding cell cycle regulatory proteins. Estrogen receptors (ERα) are intracellular receptors of estrogen. Estrogen-receptor complexes bind to estrogen response elements (ERE) in enhancers. Estrogenic control of the cell cycle occurs by increasing the expression of *cyclin* and *myc C* which promote cell cycle progression.

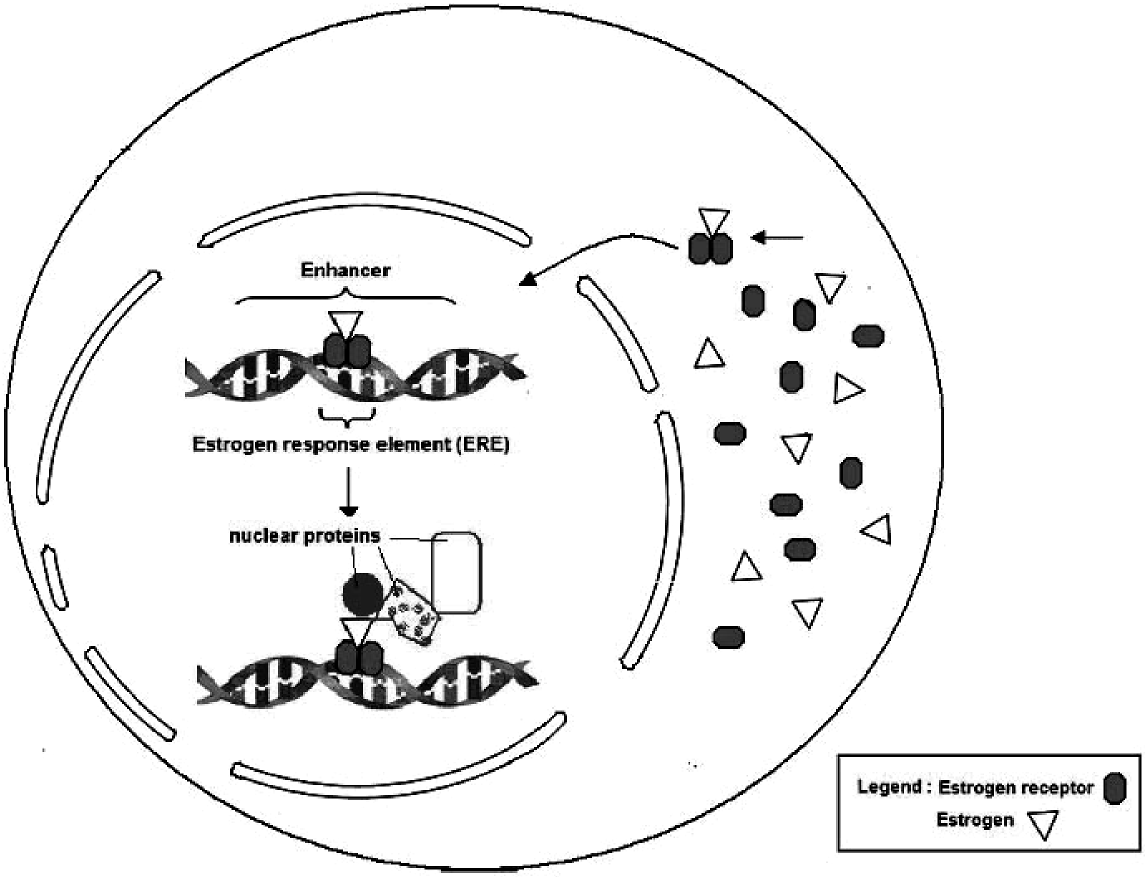

Fig 4.1

(a) With reference to the **Fig 4.1**,

(i) State the level at which gene expression is regulated by estrogen;

transcriptional level;

[1]

(ii) describe and explain how estrogen regulates gene expression;

ER in cytoplasm, signal **binds to receptors** and causes dimerisation;

enters nucleus and binds to ERE on the enhancer sequence,

attracting other transcription factors to bind; @ nuclear proteins

cause looping of DNA to **stabilize RNA polymerase/ transcription machinery;**

increases rate of transcription; @ increase / enhance gene expression

[4]

There are two isoforms of estrogen receptors: ERα and ERβ encoded by different genes. Genomic DNA samples taken from women suffering from breast cancer were analysed for the gene copy number present for the ERα gene.

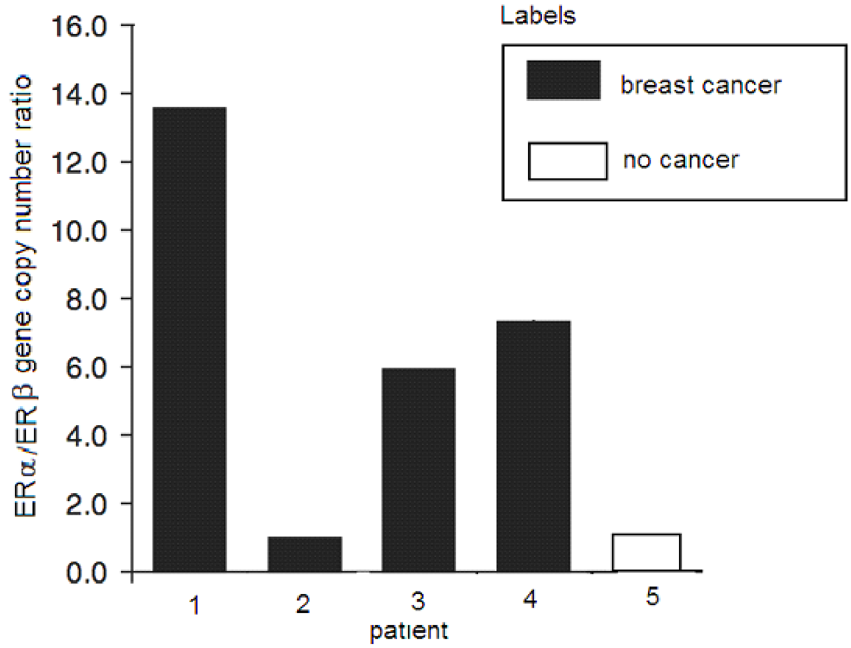

Fig 4.2

(b) With reference to the **Fig 4.2**, describe the effect different ERα gene copy number on the women.

Women 1,3, 4 who had 14 copies, 6 copies and 7 copies of ERα gene suffered from breast cancer;

Woman 2 and 5 had 1 copy of the gene but woman 2 had cancer while woman 5 did not suffer from cancer;

@ high gene copy number ratio is correlated to occurrence of breast cancer

[2]

(c) Suggest an explanation as to why patient **2** developed cancer despite a low copy number.

Cancer probably due to **mutations** on **other** protooncogenes and tumour suppressor genes**/** different mutation on ERα gene resulting in permanent activation;

@ activated promoter

[1]

(d) Suggest how amplification of the ER gene contributes to breast cancer.

more intracellular **receptors** are synthesized due to increased number of copies;

during the menstrual cycle when estrogen levels is high;

more activators bind to **increase** rate of transcription of genes encoding proteins which trigger cell division/ proto-oncogenes;

[2]

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[Total: 10]

5 Coat colour in cats is determined by a sex-linked gene with two alleles, black and ginger. When black cats are mated with ginger cats, the female offspring are always tortoiseshell, their coats showing a mottling of small black and ginger patches, while the male offspring always have the same coat colour as their mothers.

(a) Using suitable symbols, construct genetic diagrams to explain these results.

Let **X^B** represent the X chromosome carrying the **Black colour allele**
Let **X^G** represent the X chromosome carrying the **Ginger colour allele**
Let Y represent the Y chromosome (1m for legend)

Parental generation

Parental phenotypes	Black Male	x	Ginger Female
Parental genotypes	X^BY	1m: correct parental genotype	X^GX^G
After meiosis, gametes produced	X^B Y		X^G
Random fertilization			
Offspring genotype	X^BX^G	:	X^GY
Offspring Phenotype	Tortoiseshell Female	:	Ginger Male

Parental generation

Parental phenotypes	Ginger Male	x	Black Female
Parental genotypes	X^GY	1m: correct parental genotype	X^BX^B
After meiosis, gametes produced	X^G Y		X^B
Random fertilization			
Offspring genotype	X^BX^G	:	X^BY
Offspring Phenotype	Tortoiseshell Female	:	Black Male

2m: correct offspring genotype shown in both crosses
1m: both crosses showing "Tortoiseshell female and male following mother's trait"

[6]

(b) In a litter of kittens, half the number of females were tortoiseshell and the other half were black; half the number of males were black and the other half were ginger. State and explain what the colours of the parental male and female were.

X^BY x X^BX^G / Male is black, female is tortoiseshell;

male is the heterogametic sex/ male carries only 1 X chromosome/ males receive X chromosome from mother;

Half male ginger, half male black, therefore mother must have both X^B and X^G alleles;

Half female offsprings are black, therefore 1 allele must come from father, father is black/ X^BY;

[4]

[Total: 10]

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6 **(a)** State the roles of glycoproteins, carrier proteins and cholesterol in the cell surface membrane of an animal cell.

glycoproteins cell surface receptors for hormones/ neurotransmitters;

antigens;

Cell-cell adhesion; recognition; communication;

carrier proteins protein pumps in **active transport**, using ATP;

Transporting/transport polar substances against a conc gradient;

facilitated diffusion of polar substances passively;

Regulate/facilitate movement of molecules

cholesterol **maintenance/ regulation of fluidity** of membranes;

preventing the leakage of ions/ polar molecules;

[4]

Fig 6.1 shows an animal cell indicating the concentrations and direction of movement of an ion **(A)** and a non-polar molecule **(B)** on either side of the cell surface membrane.

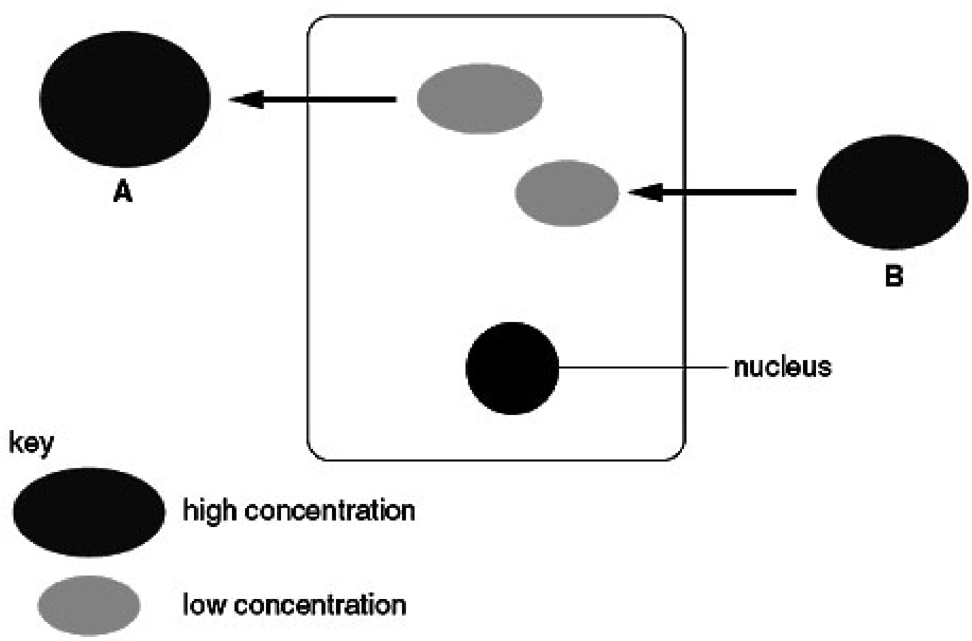

Fig 6.1

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(a) With reference to **Fig 6.1**, explain how **A** and **B** cross the cell surface membrane of the cell.

A Carrier proteins/ protein pump/ transport proteins;

moved against concentration gradient + ATP required/ active transport;

B **Moves through temporary gaps** in lipid bilayer/ **moves through hydrophobic core** of membrane;

simple diffusion/ passive transport;

[4]

(b) Describe how particles, such as bacteria, are taken up by phagocytes.

Endocytosis/phagocytosis;

receptors on cell surface/ bacteria marked by antibodies/ recognition of bacteria;

pseudopodia/ extension of cytoplasm + surrounding/enclosing/enveloping/ engulfing the bacteria + formation of vacuole/ phagosome; (at least 2 points)

[2]

(c) Phagocytes contain many lysosomes. State the function of lysosomes in phagocytes.

hydrolytic **enzymes** found in lysosome digests/ breaks down bacteria;

[1]

[Total: 11]

7 In a recent scientific expedition to study a vulnerable species of the cat family (*Felidae*), it was found that this species seemed to be heading for extinction. This species once spanned the globe, but its range is now limited to a few pockets in Africa. It was suggested that the species had somehow lost its genetic variation. To verify this suggestion, electrophoresis was used to examine the genetic variability in populations of this species. The variability was very low when compared with most populations of other mammals. If one were to select two individuals of this species at random, the chance that both being genetically very similar would be very high.

(a) Explain the following terms:

(i) species;

any **group** of organism**s** the is capable of **interbreeding** with one another;

produce **viable, fertile offspring**;

show **similarities** in evolution (phylogeny), ecology, morphology, behaviour, biochemistry;

[2]

(ii) genetic variation.

members of a **population** **differing** in one or more **inherited characteristics**;

[1]

(b) Explain the impact that low genetic variability in a population of this species has on species survival.

low number of alleles would **reduce probability** of presence of a **favourable characteristic** in the population;

upon an environmental **change**, chances of survival is reduced/ chances of species extinction in increased;

continuous inbreeding between closely related individuals leading inbreeding depression/ increase chances of recessive traits being observed/ lowers viability;

[2]

In the wild, this feline species possessed impressive skills both as a runner and hunter. However, it was discovered that there was variation in the number of spontaneous movement (i.e. tweaking movement) which occurred when the different individuals of the species run. The higher the amount of spontaneous movement, the slower the running speed of the individuals. The species was placed in standardized laboratory conditions and the number of spontaneous movement (recorded as activity score) was recorded. The results are shown in **Fig 7.1**.

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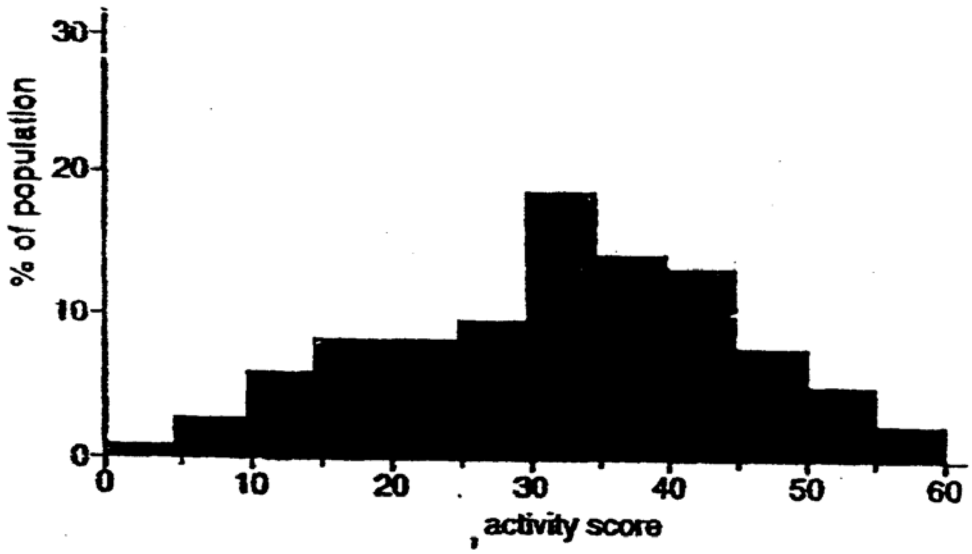

Fig. 7.1

(c) Using the information from **Fig. 7.1**, explain the type of variation shown.

continuous variation/ normal distribution + activity scores range from 0 to 60;

due to additive effects of polygenes ***OR*** more than one gene controlling the trait ***OR*** effected of environment;

[2]

The scientists then took the individuals with a low activity score and allowed them to breed among themselves. They did the same with those individuals showing high activity score. The result after 20 generations is shown in **Fig. 7.2**.

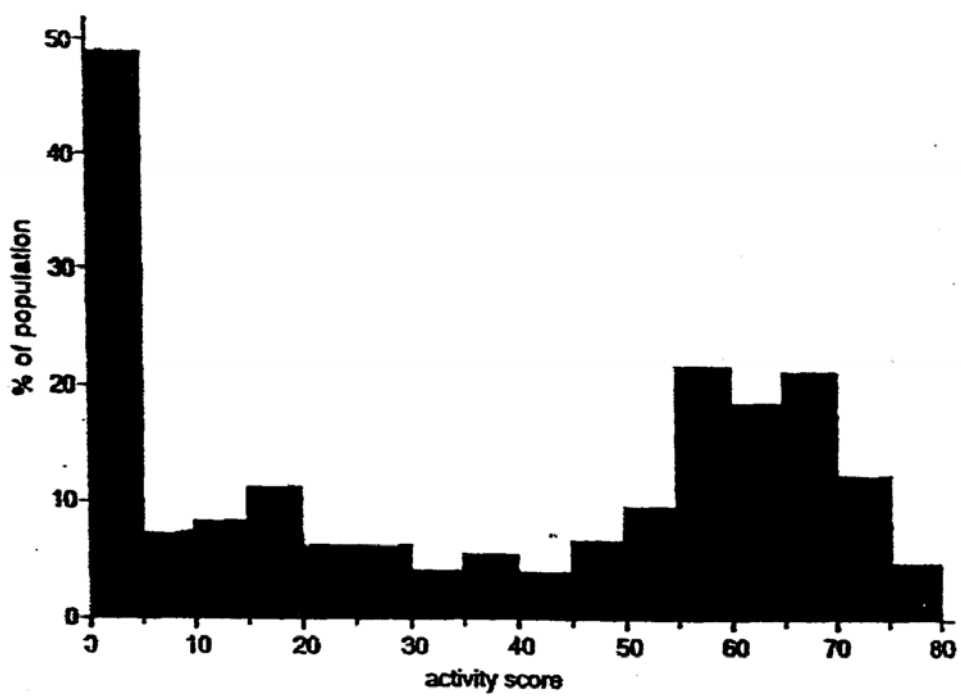

Fig. 7.2

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(d) With reference to **Fig. 7.2**,

(i) state the type of selection.

disruptive selection;

[1]

(ii) suggest what may happen if the high activity score species produce by the breeding experiment are now introduced to the low activity score species.

failure to interbreed / unable to produce fertile offspring;

competition between both species + survival of the stronger species;

[2]

(e) Being hunters, aggression is a trait observed in this feline species.

(i) Suggest how aggression probably evolved in this species.

1. individuals within population show variations in behaviour (i.e. peaceful and aggressive);

2. aggression is selectively advantageous ***OR*** aggressiveness result in increased ability to catch prey, fight off intruders ***OR*** aggressive behaviour is selectively advantageous as it is favoured by females of the species (sexual selection);

3. selected individuals survive to reproductive maturity, reproduce and pass on favourable traits to their offspring;

4. Over time, aggressive felines differ significantly from original population;

[2]

(ii) Explain why the feline population is the smallest unit that allows for the evolution of aggression.

1. require a group so variation is present

2. enough cats in group to allow interbreeding

3. NS/ selection pressure can act + some cats survive + interbreed to pass on aggression genes to the next generation;

4. changes in allelic frequencies of aggressive & non-aggressive alleles in a population;

5. Over time/ After many generations, behaviour becomes more aggressive;

[3]

[Total: 15]

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Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8 **(a)** Describe the structural features of **two** named nucleic acids involved in protein synthesis. [6]

(b) Compare the two processes that have to take place in order for protein synthesis to occur in prokaryotes. [8]

(c) Outline how the rate of protein synthesis in eukaryotes may be controlled. [6]

9 **(a)** Explain how the light independent reaction is important in photosynthesis. [8]

(b) Describe how Calvin cycle differs from Krebs cycle. [6]

(c) State the similarities between ATP production in mitochondria and chloroplasts and suggest why these similarities exist. [6]

8(a) Describe the structural features of <u>two</u> named nucleic acids involved in protein synthesis. [6]

1. named e.g. (deoxyribonucleic acid/ ribonucleic acid (tRNA/ mRNA/ rRNA) *any 2*;

<u>DNA:</u>

2. 4 types – adenine, thymine, cytosine and guanine;
3. double stranded polynucleotide + nucleotides are linked by phosphodiester/ phosphoester bonds, forming sugar phosphate backbone;
4. 2 strands held together by hydrogen bonds between complementary bases/ 2 hydrogen bonds between A & T and 3 hydrogen bonds between G & C;
5. bases are stacked above one another at a constant distance of 3.4nm apart;
6. hydrophobic interactions between stacked bases;
7. 2 polynucleotide strands are anti-parrellel/ one poly nucleotide strand runs in the 3' to 5' direction while the other runs in the 5' to 3' direction;
8. helical structure;
9. 10 nucleotides per helix with a length of 3.4nm;
10. sugar-phosphate backbone found outside the helix, hydrophobic nitrogenous bases on the inside/ core of helix;
11. major and minor grooves;

mRNA:

12. 4 types of ribonucleotides + adenine, cytosine, guanine, uracil;
13. single stranded molecule held together by phosphodiester bond;
14. 5' guanine cap and 3' poly-A tail
15. exons separated by introns;
16. 5' and 3' untranslated regions;
17. random coil;

tRNA:

18. 4 types of ribonucleotides + adenine, cytosine, guanine, uracil;

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19. single stranded molecule held together by phosphodiester bond;
20. 2D structure is a cloverleaf structure formed by complementary base pairing;
21. 3D structure is a compact L-shaped structure containing an anticodon in one of the loops
22. 3' –OH (CCA) binding site for amino acids;

rRNA:
23. nucleolus
24. assembled with ribosomal proteins;

8(b) Compare the two processes that have to take place in order for protein synthesis to occur in prokaryotes..[8]

1. both involve **complementary base pairing;**
2. Both take place **simultaneously in the cytoplasm** ;

Feature	***Transcription***	***Translation***
3. **mode of termination**	**Rho dependent / independent proteins**	**Stop codons**
4. mode of initiation	**Pribnow box (promoter)**	**Start codon AUG**
5. Monomers	**ribonucleotides**	**Amino acids**
6. **Involvement of ribosome**	**not involved**	**Ribosome is the binding site for mRNA** and **amino-acid tRNA complex**.
7. **Template**	**DNA template/ coding strand**	**mRNA**
8. **Enzyme**	**RNA polymerase catalyses formation of phosphodiester bond** between adjacent ribonucleotides	**peptidyl transferase** catalyzes the **formation of peptide bond**.
9. **Bond between basic units.**	**Bond is formed** between the **phosphate group of one nucleotide** and **carbon atom 3 of ribose** of the next nucleotide.	**Bond is formed** between **carboxyl group of one amino acid** and the **amino group of the next amino acid**.
10. **reading of genetic message**	**RNA polymerase moves along DNA template/ coding strand.**	**Ribosome moves along mRNA.**
11. **Involvement of tRNA**	**not involved**.	**tRNA carries amino acids** to ribosomes.
12. **raw material(s)**	**free ribonucleotides**	**amino acids attached to tRNA**
13. **product(s)**	**mRNA**, **rRNA** and **tRNA**	**polypeptide**
14. **Fate of product(s)**	Products **exit nucleus** and **migrate to the cytoplasm**	**polypeptide chain remain in the cytoplasm** or **secreted out of the cell**

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8(c) Outline how the rate of protein synthesis in eukaryotes may be controlled.[6]

Chromatin and histone modification:

1 DNA methylation results in recruitment of histone deacetylases, which then results in the DNA being coiled more tightly, thus lowering rate of transcription;

Transcriptional control:

2 presence of basal transcription factors only lead to low/ basal transcription rates;
3 combinatorial effects of control elements such as enhancers and silencers will increase and decrease rates of transcription respectively;

Post-transcriptional control

4 RNA processing where a 5' methylguanosine cap is added;
5 3' polyA tail is added by polyA polymerase;
6 prevent or delay degradation by exonucleases/ increase stability of mRNA, thus prolonging the lifespan of the mRNAs, more protein produced per unit time;

Translational control

7 Presence of activated/ phosphorylated translation initiation factors increase stability of ribosome binding;
8 translation repressors which bind to 5' UTR, preventing ribosome from binding or proceeding thus lowering translation rate;
9 temporal control mRNA are synthesised with a short polyA tail and are protected from degradation;
10 mRNA localization results in rates of protein synthesis differing in different tissues;
11 high mRNA levels concentrate in some cells to ensure high levels of protein to be synthesised per unit time in those tissues;
12 rates of translation of these mRNA will only increase when cytoplasmic polymerase catalyses the cytoplasmic polyadenylation at specific times in during the maturity of the cell;
13 amount/ quantity/ number of ribosomes present/ reference to enhanced expression of ribosomal genes in nucleoli;
14 amount of aminoacyl tRNA transferase/ activated tRNA;
15 presence of specific tRNA means correct amino acid is correctly brought to the ribosome, thus rate of translation is enhanced;

9a Explain how the light independent reaction is important in photosynthesis. [8]

Purpose:

1. synthesis of carbohydrates: products are sucrose/ starch through reversal of glycolysis;
2. synthesis of fatty acids/ glycerol when GP is converted to an acetyl group/ glycerol synthesised from triose phosphate (TP);
3. synthesis of proteins from GP through transamination/ addition of amino group after converting GP to acetylcoA to ketoglutarate then to amino acid;

Process:

4. carbon fixation/acceptance of carbon dioxide by ribulose bisphosphate;
5. catalysed by RuBP carboxylase-oxygenase;
6. forming unstable 6C intermediate which immediately breaks down to become glycerate 3 phosphate (GP);
7. GP converted to glyceraldehyde 3 phosphate (GALP)/ TP via reducing power of $NADPH+H^+$ and ATP;
8. RUBP regenerated;

9b Describe how Calvin cycle differs from Krebs cycle. [6]

1. ***site***: stroma of chloroplast vs matrix of mitochondrion;
2. ***electron or proton carriers***: $NADP^+$ to become $NADPH+H^+$ vs NAD^+ to $NADH+H^+$ & FAD to $FADH_2$; (*spell out once)*
3. ***role of CO_2:*** accepted by **ribulose bisphosphate** in carbon fixation vs released by oxidative decarboxylation;
4. carbon fixation catalysed by **ribulose bisphosphate carboxylase oxygenase**;
5. ATP is used in the regeneration of RuBP;
6. ***role of ATP***: **used in reduction** of glycerate phosphate to glyceraldehyde phosphate vs **synthesised** via substrate level phosphorylation;
7. ***product/ substrate***: **synthesizes glucose**/ starch with the **investment of ATP** vs **breaks down glucose** to **produce ATP**;

9c State the similarities between ATP production in mitochondria and chloroplasts and suggest why these similarities exist. [6]

1. involvement of reduced coenzymes as H^+ and e^- carriers;
2. electrons transported along ETC with carriers each with an energy level lower than the one preceeding it;
3. proton/ H^+ actively pumped across membrane using energy from the flow of electrons;
4. generation of electrochemical proton gradient across membrane;
5. diffusion of protons through stalked particles;
6. stalked particles contain ATP synthase to synthesize ATP from ADP + Pi;
7. similarities because both mitochondrion and chloroplast were thought to be free-living prokaryotes which were engulfed by a larger pre-eukaryote cell;
8. symbiotic/ coexistence because of mutual benefits, endosymbiont theory;